

Further Pure Paper 1 — Claim Map

**DO NOT OPEN UNTIL THE PAPER IS FINISHED
AND YOUR SOLID / RUSTY / GAP PREDICTIONS ARE WRITTEN DOWN**

§B What each question is testing — do not read before marking

Difficulty bands: **A** = anchor (expect to get these; they confirm the floor). **B** = discriminator (this is where the A/A* gap lives — unscaffolded, method not given). **C** = stretch (above A* standard; tells you about unfamiliar problem solving, not about your grade).

Q	Band	Topic	The specific claim being tested	Attribute
1	A	Complex numbers	Modulus–argument form and de Moivre for a clean case	Fluency
2	B	3×3 inverse — logged gap, never completed	Can you carry out the full cofactor/adjugate route without losing a sign, and do you check the determinant <i>first</i> ?	Fluency + conditional
3	A	Roots of polynomials	Vieta's relations recalled correctly; the $\sum \alpha^2 = (\sum \alpha)^2 - 2 \sum \alpha\beta$ step	Fluency
4	B	Skew lines — logged gap	Do you <i>classify before computing</i> ? Many students compute a distance without establishing the lines aren't parallel or intersecting. The justification is the marks.	Method selection
5	B	Induction — logged gap (base case; assumed vs proved)	The stated assumption. §3 records the assumed-vs-proved confusion directly; this question makes it visible before the algebra hides it.	Proof and rigour
6	A	Complex loci	Recognising equidistance $\Rightarrow$ perpendicular bisector, rather than expanding blindly	Connections
7	B	Point-to-plane distance — logged gap, unscaffolded	Three chained steps (two direction vectors $\rightarrow$ normal $\rightarrow$ plane equation $\rightarrow$ distance) with no parts to guide you. Failure here could be any of four things — note which step broke.	Fluency + route choice
8	C	Roots transformations — logged gap	Squaring roots needs $\sum \alpha^2 \beta^2 = (\sum \alpha\beta)^2 - 2\alpha\beta\gamma \sum \alpha$, which is not a substitution. If you tried $y = x^2$ and substituted, that is the finding.	Generalisation
9	A	Series	Standard results applied and factorised	Fluency
10	B	Invariant lines	Do you know invariant line through origin $\Leftrightarrow$ eigenvector direction? Or did you set $y = mx$ and grind? Both work; only one shows the connection.	Connections
11	C	de Moivre $\Rightarrow$ multiple-angle identity	Unscaffolded. Normally given as "expand $(\cos \theta + i \sin \theta)^5$, hence...". Without that line, can you find the route?	Unfamiliar problem solving
12	B	Method of differences	Spotting that partial fractions telescope, and handling the two leftover terms at each end correctly — the classic slip	Fluency + recognition
13	A	2×2 inverse + interpretation	The inverse is the anchor; the two sentences are the real test — procedural fluency without geometric meaning is exactly the pattern §1 warns about	Conceptual understanding
14	B	Pure conditional knowledge	The most informative five minutes on the paper. §3 says most of your failures are conditional. This tests it with no execution to hide behind.	Method selection
15	C	Diagonalisation	Connecting Q10 to a power. Tests whether eigen-work is a procedure or a tool.	Connections

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16	—	Declarative sweep	Cheap, fast, catches definition-level holes. §3 records $\mathbb{Q}/\mathbb{R}$ swapped and injective stated backwards — definitions are a known weak spot.	Conceptual understanding

How to read your results

- **Failed an anchor (1, 3, 6, 9, 13):** treat as urgent. These are below week-5 standard.
- **Failed a discriminator (2, 4, 5, 7, 10, 12, 14):** this is the $A \rightarrow A^*$ gap, and it is the actual target for the next eleven days.
- **Failed a stretch (8, 11, 15):** informative, not urgent. Do not let these set the revision priority.
- **Q14 answered vaguely:** the single most important signal on the paper. Vague method selection under no time pressure means it will be worse under time pressure.
- **Predicted SOLID but got it wrong:** the most valuable row in the whole exercise. Miscalibration is more dangerous than a gap, because you will not revise it.

Known limits of this paper

- Built on OCR Core Pure content. **If you are on MEI, some notation differs** — the mathematics doesn't.
- It does not cover: hyperbolic functions, polar coordinates, Maclaurin series, differential equations, or any optional-paper content. If week 5 includes those, this paper does not measure them and a second short diagnostic is needed.
- It samples. One bad evening looks like a gap. Anything that reads as a gap here needs a retest at 10–14 days before it counts as real (Coaching Document §4).
- These are custom questions, not official past-paper items. Per §4 they are good for *finding* gaps but do not by themselves prove a topic solid — confirm anything that looks fine against an OCR paper.